

TOKEN BALANCING, UPDATE SPARSITY, AND SUPERVISION DENSITY IN MULTI-TEACHER ON-POLICY DISTILLATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Multi-teacher on-policy distillation combines supervision from specialized teachers through one evolving student. We study how loss averaging, optimizer state, and vocabulary supervision shape this process using an aligned Qwen3-1.7B experiment with four domain teachers. Thirteen capability evaluations show distinct early tradeoffs among three averaging rules: at update 50, global token averaging improves mathematics over domain response averaging by 4.6 percentage points, while domain token averaging reduces instruction-following accuracy by 2.0 points. Parameter measurements reveal a second distinction. After 100 sampled-gradient updates, 94.0% of FP32 master coordinates have changed, compared with 3.41% of BF16 model coordinates. At this checkpoint, saved Adam moments produce substantial movement even with a zero current gradient. Fixed-prefix comparisons further show that nearly identical writeback densities can conceal different update directions: sampled and full-vocabulary gradients have mean cosine 0.677, while their BF16 writeback steps have cosine 0.915. Together, these results connect capability allocation to the quantities actually averaged, accumulated, and stored during distillation, and identify input composition and optimizer history as essential variables in interpreting teacher-conditioned parameter geometry.

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) trains a shared student on its own responses using feedback from domain specialists (Ma et al., 2026). Its appeal is direct: mathematics, code, instruction following, and science can share one model while retaining specialized supervision. Its optimization is less transparent. Equal prompt quotas can yield unequal token weights, different teachers can act on overlapping coordinates, and a dense loss can leave only a small visible change in stored weights.

These observations concern three successive transformations. Loss averaging determines how responses contribute to the gradient. Adam combines that gradient with accumulated state. Finite-precision writeback determines which changes appear in the model checkpoint. Understanding the resulting capability allocation requires measuring these transformations separately. Token-share balancing already motivates interventions in Open-MOPD (Gao et al., 2026), while recent RL and distillation studies document sparse parameter changes and investigate restricted parameter updates (Mukherjee et al., 2025; Yu et al., 2026). We examine how these phenomena interact in a common multi-teacher setting.

Our study uses a Qwen3-1.7B student initialized from the common parent of four RL teachers, with shared prompt formatting, decoding conventions, and optimizer settings. Six training configurations provide thirteen completed capability suites. Measurements from the joint sampled-gradient checkpoint at update 100 connect these trajectories to teacher-conditioned gradients and controlled local Adam steps. Two fixed response banks support comparisons of input domains, teacher identities, and vocabulary supervision.

Three findings organize the paper. First, averaging rules induce different early capability tradeoffs; fixing domain weights and fixing response weights are separate operations (Section 3). Second, the apparent sparsity of a parameter change depends strongly on precision and optimizer history, and

teacher-conditioned gradient directions depend on input composition (Section 4). Third, writeback density can remain nearly constant while the direction changes with the supervision estimator (Section 5). These findings provide a concrete account of how supervision is converted into parameter movement during multi-teacher distillation.

2 SETTING AND OBJECTIVES

Let $p_\theta(v \mid h)$ be the student distribution and $q_d(v \mid h)$ the teacher for domain d , at prefix $h = (x, y_{<t})$. A fresh response y is sampled from the student for each training prompt. Teachers score the same student prefix. We write $\text{sg}[\cdot]$ for stop-gradient and suppress h below.

Sampled supervision (PG). For sampled token a , the implemented loss is

$$\ell_{\text{PG}}(a) = -\text{sg}[\log q_d(a) - \log p_\theta(a)] \log p_\theta(a). \quad (1)$$

The coefficient is detached, with no advantage or policy-ratio clipping. At a fixed prefix, fresh actions $a \sim p_\theta$ give an unbiased gradient of the full-vocabulary reverse KL,

$$\ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_\theta(v) \log \frac{p_\theta(v)}{q_d(v)}. \quad (2)$$

This identity supplies the reference for the fixed-prefix sampling experiment in Section 5.

Intersection supervision (I64). Let S be the rollout student’s top-64 set, T the teacher’s top-64 set, and $I = S \cap T$. The online I64 loss is

$$\ell_{\text{I64}} = -\sum_{v \in I} \text{sg} \left[\frac{p_\theta(v)}{\sum_{u \in S} p_\theta(u)} (\log q_d(v) - \log p_\theta(v)) \right] \log p_\theta(v). \quad (3)$$

Probabilities use full-vocabulary normalization. The student weights are normalized over the original S and then masked to I ; actor probabilities are refreshed at the saved candidate positions. An empty intersection contributes zero. The local diagnostic also evaluates a corrected teacher-top-64 loss (T64), defined in Appendix D.

Experimental configuration. M-PG uses four routed teachers and domain response averaging. M-I64-DR, M-I64-DT, and M-I64-GT combine I64 with domain response, domain token, and global token averaging, respectively. S-PG and S-I64 use the mathematics teacher and mathematics prompts. Each optimizer update consumes 64 responses: 16 per domain in the multi-teacher runs and 64 mathematics responses in the single-teacher runs. Training uses seed 42 and a 4,096-token response cap. The capability results cover 22,360 responses across thirteen complete evaluations; full settings and checkpoint coverage appear in Appendix A.

3 TOKEN BALANCING AND CAPABILITY ALLOCATION

3.1 DOMAIN WEIGHTS AND RESPONSE WEIGHTS

For response i , let T_i be its valid-token count and $\ell_i = T_i^{-1} \sum_t \ell_{i,t}$ its mean token loss. Let $\mathcal{B}_d$ contain the responses from domain d , with target weights $\sum_d \lambda_d = 1$. The three reductions in Table 1 expose two distinct choices: the weight assigned to a domain, and the weight assigned to responses within that domain.

With equal prompt quotas, longer responses increase a domain’s weight under GT. DT removes this between-domain length effect, while retaining length weighting within each domain. Writing $g_i = \nabla_\theta \ell_i$, the exact fixed-batch relation is

$$\frac{\sum_{i \in \mathcal{B}_d} T_i g_i}{\sum_{i \in \mathcal{B}_d} T_i} = \bar{g}_d + \frac{\text{Cov}_d(T, g)}{\bar{T}_d}. \quad (4)$$

Thus the effect of DT relative to DR depends on the covariance between response length and gradient, even after domain weights are fixed. Appendix B derives the identity. In the first 50 rollouts

Table 1: Averaging rules on a fixed batch of nonempty responses. We use $\lambda_d = 1/4$. GT assigns domains their observed token shares; DT fixes domain weights; DR also gives each response equal weight within its domain.

Reduction	Batch loss
Global token (GT)	$\sum_i T_i \ell_i / \sum_i T_i$
Domain token (DT)	$\sum_d \lambda_d \sum_{i \in \mathcal{B}_d} T_i \ell_i / \sum_{i \in \mathcal{B}_d} T_i$
Domain response (DR)	$\sum_d \lambda_d \mathcal{B}_d ^{-1} \sum_{i \in \mathcal{B}_d} \ell_i$

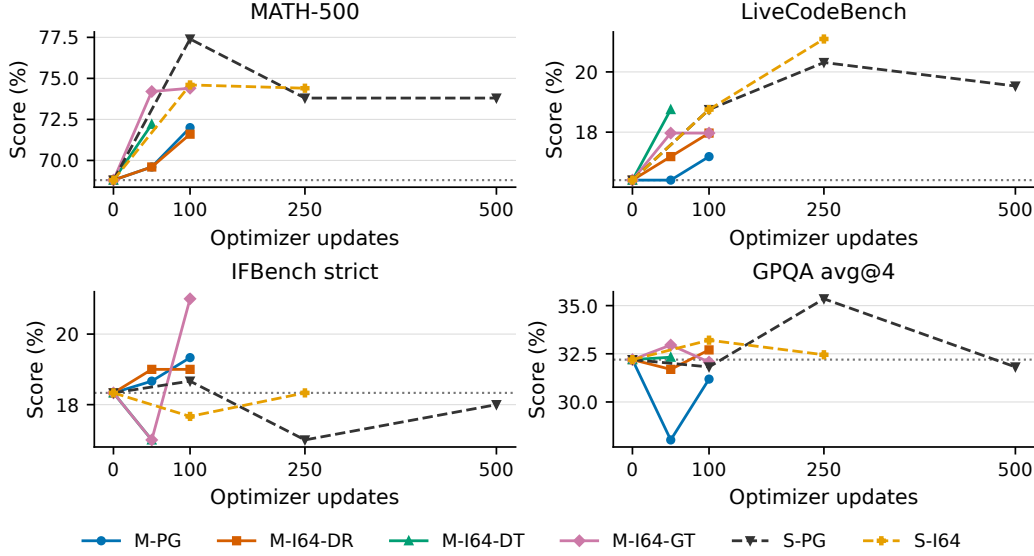


Figure 1: Capability trajectories from the shared initial student. Markers are completed evaluations; lines connect measured checkpoints. M denotes four routed teachers and S the mathematics teacher. GPQA reports average accuracy over four responses per question. All curves use one training seed; checkpoint availability is recorded in Table 3.

of M-I64-GT, mathematics receives 44.05% of valid tokens and instruction following 14.98%, averaged over rollout batches, despite both receiving 25% of prompts. Under GT these are also their mean loss weights; DT and DR assign each domain 25%. The token-share and response-length traces in Appendix F show how these weights evolve.

3.2 EARLY CAPABILITY TRADEOFFS

Figure 1 shows every completed evaluation in the aligned campaign. M-PG reaches 72.0% on mathematics at update 100, up from 68.8% initially; instruction following rises from 18.33% to 19.33%, while GPQA moves from 32.20% to 31.19%. The trajectories therefore reflect domain-specific changes rather than a common gain across capabilities. The single-teacher curves provide a mathematics-specialization reference, with four times as many mathematics prompts per update.

The three I64 reductions can be compared at their shared update-50 checkpoint (Table 2). GT leads mathematics, while DR leads instruction following. Relative to DR, GT gains 4.6 percentage points on mathematics, with a paired prompt-bootstrap interval of $[1.2, 8.2]$ points. DT changes instruction following by -2.0 points, with interval $[-4.0, -0.33]$. Figure 6 gives all eight paired contrasts and the uncertainty procedure. These results make the choice of averaging rule a capability-allocation decision: enforcing equal domain mass changes which responses drive learning, and its value depends on the target capability.

Table 2: Capability scores (%) at the common comparison checkpoint. Initialization, teachers, prompt quotas, and optimizer settings are shared. Each branch generates responses from its evolving student.

I64 reduction, update 50	Math	Code	IF	GPQA
DR	69.60	17.19	19.00	31.69
DT	72.20	18.75	17.00	32.32
GT	74.20	17.97	17.00	32.95

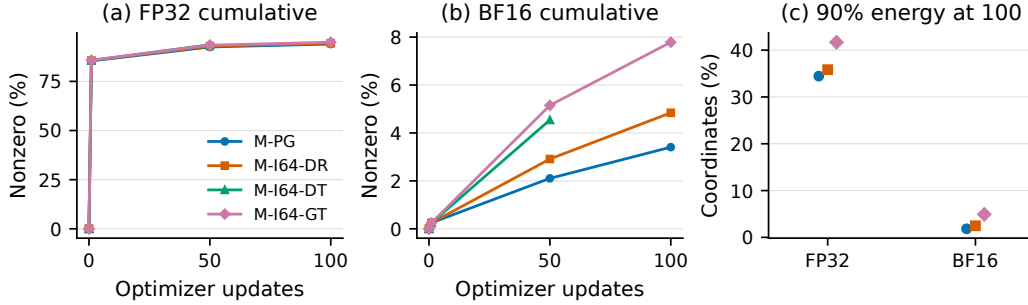


Figure 2: Actual online parameter changes from initialization. (a,b) Exact nonzero fractions in FP32 master and BF16 model coordinates. (c) The coordinate fraction containing 90% of squared cumulative change at update 100. Precision changes both measured support and energy concentration. DT is available through update 50.

4 FROM GRADIENTS TO STORED PARAMETER CHANGES

4.1 PRECISION CHANGES THE MEASURED SUPPORT

For a parameter change $\Delta \in \mathbb{R}^P$, define its active fraction at threshold τ as $f_\tau(\Delta) = P^{-1} \sum_j \mathbf{1}\{|\Delta_j| > \tau\}$. We measure both the cumulative change from initialization and individual optimizer steps, using the FP32 master weights and the BF16 model weights. The coordinate system contains 1,720,574,976 parameters after removing vocabulary padding and shared duplicates.

At update 100, the M-PG cumulative FP32 change has 94.0% nonzero coordinates; the BF16 change has 3.41% (Figure 2). The smallest sets containing 90% of squared change contain 34.46% and 1.82% of coordinates, respectively. The actual FP32 optimizer step at that checkpoint is also dense, with 88.13% nonzero coordinates. These measurements locate the strongest sparsity in the stored BF16 change. The reductions produce different trajectories: DR and GT reach 4.84% and 7.79% BF16 cumulative activity at update 100, alongside different total change norms.

4.2 SAVED ADAM STATE EXPLAINS MOVEMENT BEYOND THE CURRENT GRADIENT

We examine the M-PG checkpoint at update 100 using two response banks. Each bank contains one student response per domain and four selected prefixes per response, giving 32 prefixes in total. Student and teacher forward/backward computations use FP32 at the saved BF16 weight values, with TF32 disabled. A local non-fused Adam calculation then uses the checkpoint’s FP32 master weights and moments. This isolates the transformation from a fixed-bank gradient to a proposed parameter step.

Figure 3 compares the saved-state calculation with two state interventions. With zero current gradient, Adam changes 0.1403% of BF16 coordinates. With the sampled PG gradient, it changes 0.1486%. Resetting the first moment reduces the PG fraction to 0.0504%; resetting both moments and the step clock gives 0.5140%. Stored momentum and the adaptive denominator therefore affect both the size and location of visible writeback. The zero-gradient branch also has cosine 0.901 with the PG BF16 step. A no-step cast exactly reproduces the saved model, establishing that this movement arises during the proposed Adam step.

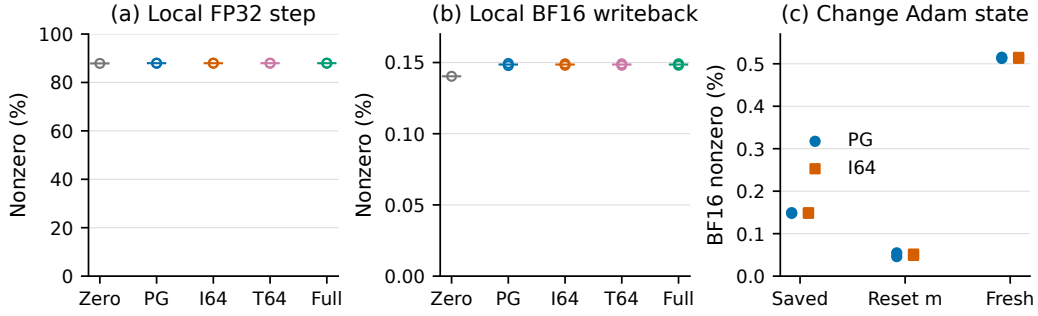


Figure 3: Local Adam calculations at the same saved state. Points show the two response banks; the zero-gradient control is evaluated once. (a,b) Nonzero fractions for FP32 steps and BF16 writeback. (c) PG and I64 with saved moments, reset first moment, or fresh moments and clock. Weight decay is zero. These state interventions reveal how optimizer history and rounding shape visible support.

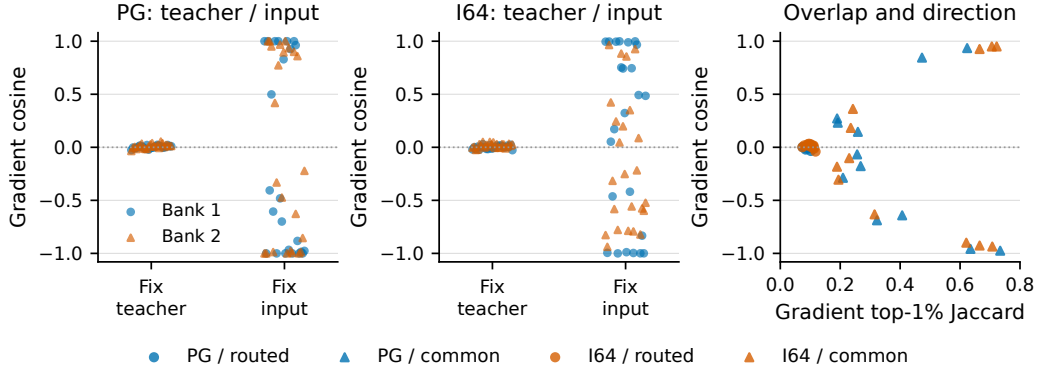


Figure 4: Teacher-conditioned gradient geometry at M-PG update 100. Left and center: crossed comparisons with four prefixes per gradient, grouped by which factor is held fixed. Each dot is a dependent pair comparison within one of two banks. Right: largest-magnitude 1% coordinate overlap versus cosine for routed and common-bank gradients. Appendix C records norms and tie handling; Appendix E relates overlap to teacher-distribution distance.

4.3 INPUT COMPOSITION SEPARATES PARAMETER OVERLAP FROM DIRECTION

The same checkpoint supports a crossed comparison of four input domains and four teachers. For each bank, every teacher scores the same four prefixes from each domain response. Holding the teacher fixed while changing the input domain yields mean gradient cosines of 0.0059 for PG and 0.0072 for I64. Holding the input domain fixed while changing the teacher produces both strongly aligned and strongly opposed gradients (Figure 4). Input composition is therefore a substantial part of the geometry attributed to a routed teacher.

For the routed comparison, each teacher uses its own domain response. Mean teacher-pair cosines are -0.0020 for PG and 0.0081 for I64, while the mean Jaccard overlaps of their largest-magnitude 1% gradient coordinates are 0.0951 and 0.0999. Averaging each teacher across the common four-response bank raises these overlaps to 0.3808 and 0.4577. The associated cosines still span negative and positive values. The common-bank comparison changes both input composition and response count; the crossed comparison fixes response count and exposes the dependence on domain and teacher separately. Parameter locations and signed directions carry different information about teacher interaction.

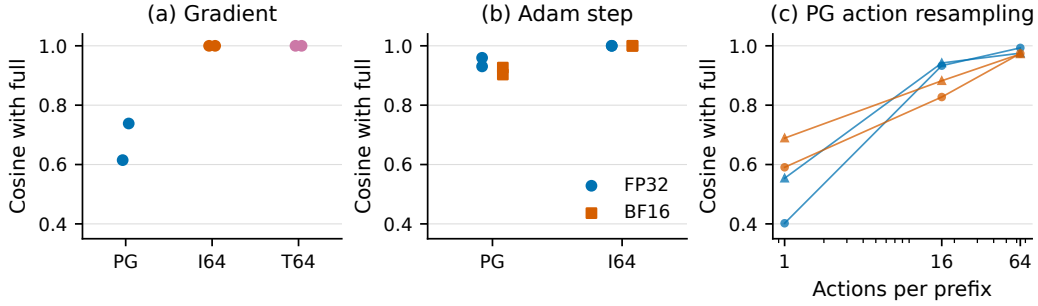


Figure 5: Supervision comparisons on fixed prefixes at update 100. (a) Original-bank gradient cosine with full-vocabulary reverse KL. (b) Cosine of local Adam steps computed with the same saved state. (c) Fresh PG actions at each fixed prefix, with two banks and two independent draws per sample count. Colors distinguish banks and marker shapes distinguish repeats in (c); counts use separate draws. All panels compare direction.

5 SUPERVISION DENSITY CHANGES DIRECTION MORE THAN WRITEBACK DENSITY

The fixed-prefix banks also compare sampled PG, I64, corrected teacher-top-64 (T64), and full-vocabulary reverse KL. Averaged across banks, the PG gradient has cosine 0.6766 with the full-vocabulary gradient. I64 and T64 reach 0.99999990 and 0.99999981 (Figure 5a). This agreement occurs on prefixes with concentrated student probability: averaged within each bank-domain group, the student places 99.9184–100% of its probability on the teacher’s top-64 tokens. Appendix D reports the coverage and local entropy measurements.

Adam and writeback retain part of the directional difference while making coordinate counts nearly identical. The BF16 active fractions are 0.148599% for PG and 0.148587% for both I64 and full vocabulary. Yet PG/full step cosine is 0.9451 in FP32 and 0.9149 in BF16; I64/full reaches 0.999998 and 0.999956. Thus a single density statistic can hide a measurable change in which direction the optimizer moves.

To test the role of action sampling, we resample fresh actions at each fixed prefix and average 1, 16, or 64 PG contributions. Mean gradient cosine with the exact reference rises from 0.5589 to 0.8962 and 0.9787, while relative norm error falls from 1.4836 to 0.5148 and 0.2221 (Figure 5c). Holding prefixes and weights fixed makes this a direct comparison of realized sampling error. The original-bank and resampled one-action results use different actions, accounting for their different starting cosines.

The capability trajectories provide the corresponding online comparison. At update 100, S-PG and S-I64 score 77.4% and 74.6% on mathematics; at update 250 they score 73.8% and 74.4%. At update 500, S-PG remains at 73.8% on mathematics, with code at 19.53%, instruction following at 18.0%, and GPQA at 31.82%. Among evaluated checkpoints, S-PG peaks on mathematics at update 100 and retains a 5.0-point gain over initialization at update 500. The fixed-prefix measurements characterize the estimator and Adam transformation at a given state; the capability curves track the resulting trained models.

6 RELATED WORK

MOPD combines domain-specialized teachers by scoring student-generated responses, using policy-gradient and distribution-based supervision (Ma et al., 2026). Open-MOPD studies capability imbalance in this setting and introduces interventions for task weighting and response behavior (Gao et al., 2026). Our normalization comparison connects the choice of averaging denominator to aligned capability trajectories and recorded token allocation.

Sparse parameter changes have also been studied in reinforcement learning and distillation. Mukherjee et al. (2025) investigate small parameter subsets that support RL fine-tuning. Yu et al. (2026) study sparse changes under dense supervision and evaluate restricted parameter updates. We connect this line of work to multi-teacher gradient geometry by measuring FP32 and BF16 changes, intervening on saved Adam state, and crossing input domains with teacher identities. These measurements explain how gradient support, update direction, and stored parameter activity can exhibit different patterns within the same student.

7 DISCUSSION

The experiments identify three decisions that shape multi-teacher distillation: which responses receive loss weight, how their gradients combine with optimizer history, and which increments survive model writeback. At a common early checkpoint, the averaging rule changes the capability trade-off. At a fixed student state, input composition changes teacher-conditioned directions, and broader vocabulary supervision changes gradient direction even when BF16 activity remains nearly constant. Reporting these quantities together makes parameter geometry more informative about the optimization process.

The present evidence covers one model family, four related teachers, one training seed, and local calculations at update 100. The two diagnostic banks contain eight responses generated with a 256-token cap; all reach that cap, and the selected prefixes have high top-64 coverage. Extending the analysis to later and higher-entropy prefixes would test how this coverage affects the supervision comparison. Completing the aligned checkpoint grid and adding training seeds would establish the persistence of the capability tradeoffs. Restricted-parameter training would connect the measured gradient supports to operational subnetworks.

REPRODUCIBILITY STATEMENT

The accompanying evidence bundle contains the values used in every figure and table, source hashes, configuration records, a scoring audit, and scripts that regenerate the plots on CPU. Eight capability suites are verified against response-level artifacts; four single-teacher suites use previously verified summaries from a second node. Appendix A gives the evaluation protocol, checkpoint coverage, and uncertainty procedure. Appendix C specifies the parameter coordinates and local optimizer calculations.

ETHICS STATEMENT

The experiments use public model families and benchmark datasets to study optimization and capability transfer. We report standard benchmark scores together with response completion and repetition diagnostics. Generative AI assisted manuscript editing and analysis code; the authors are responsible for the reported measurements and their interpretation.

REFERENCES

- Huan-ang Gao, Haohan Chi, Yong Yan, Shiyuan Feng, Hanlin Wu, Zheng Jiang, Bingxiang He, Wei-Ying Ma, Ya-Qin Zhang, and Hao Zhou. Open-MOPD: Diagnosing and fixing capability imbalance in multi-teacher on-policy distillation, 2026. URL <https://arxiv.org/abs/2608.19098>.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2103.03874>.
- Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code, 2024. URL <https://arxiv.org/abs/2403.07974>.

- Wenhan Ma, Jianyu Wei, Liang Zhao, Hailin Zhang, Bangjun Xiao, Lei Li, Qibin Yang, Bofei Gao, Yudong Wang, Rang Li, Jinhao Dong, Zhifang Sui, and Fuli Luo. MOPD: Multi-teacher on-policy distillation for capability integration in llm post-training, 2026. URL <https://arxiv.org/abs/2606.30406>.
- Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tur, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models, 2025. URL <https://arxiv.org/abs/2505.11711v2>. Version 2; NeurIPS 2025.
- Valentina Pyatkin, Saumya Malik, Victoria Graf, Hamish Ivison, Shengyi Huang, Pradeep Dasigi, Nathan Lambert, and Hannaneh Hajishirzi. Generalizing verifiable instruction following, 2025. URL <https://arxiv.org/abs/2507.02833>.
- Qwen Team. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2023. URL <https://arxiv.org/abs/2305.18290>.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof q&a benchmark, 2023. URL <https://arxiv.org/abs/2311.12022>.
- Guo Yu, Wenlin Liu, Yulan Hu, Hao-Xuan Ma, Jun-Peng Jiang, and Han-Jia Ye. Dense supervision, sparse updates: On the sparsity and geometry of on-policy distillation, 2026. URL <https://arxiv.org/abs/2606.13657v3>. Version 3, July 30, 2026.

A EXPERIMENTAL SETUP AND COMPLETE CAPABILITY RESULTS

A.1 MODEL, DATA, AND TRAINING PROTOCOL

The student is Qwen3-1.7B (Qwen Team, 2025), revision 70d244cc86cc (the full revision and weight hashes are in the frozen protocol). The four frozen teachers are mathematics, code, instruction-following, and science RL descendants of the same initial model. Student and teacher use the native chat template with thinking disabled, the same assistant prefix, and no suffix removal. Generation stops at token 151643 or 151645; the model EOS token is 151645. The vocabulary contains 151,936 entries.

All six configurations use AdamW with learning rate 2.5×10^{-7} , $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-8}$, zero weight decay, constant learning rate, and global gradient clipping at norm 1. Training sampling uses temperature 1 and top- $p = 1$, with a 2,048-token prompt cap and 4,096-token response cap. One fresh response is sampled per prompt. An optimizer update aggregates 64 responses in sixteen four-response slices, with one response per execution microbatch. The optimizer is applied once after all domain blocks are accumulated. Consequently, the scalar logger’s `train/step` counts gradient slices; all checkpoint axes in this paper count actual optimizer updates.

The multi-teacher configurations allocate 16 prompts to each domain per update and use target domain weights $1/4$. Single-teacher configurations allocate all 64 prompts to mathematics. Prompts follow a seeded shuffled schedule, repeated after exhaustion. The frozen protocol records the exact train, diagnostic, and held-out split hashes and teacher conversion manifests. I64 starts from initialization with the intersection objective in Equation 3.

Table 3: Coverage of the aligned evidence snapshot. The single-teacher training runs reached update 500; complete capability evaluations reach 500 for S-PG and 250 for S-I64. Multi-teacher runs have unequal coverage after interruptions and restarts. The DT update-100 checkpoint is incomplete. Reported curves use the completed measurements listed here.

Configuration	Complete capability checkpoints	Online geometry in this paper
Initial	0	Initial reference
S-PG	100, 250, 500	—
S-I64	100, 250	—
M-PG	50, 100	0, 1, 50, 100; local probes at 100
M-I64-DR	50, 100	0, 1, 50, 100
M-I64-DT	50	0, 1, 50
M-I64-GT	50, 100	0, 1, 50, 100

A.2 EVALUATION AND SCORING

Each capability suite contains MATH-500 (500 questions) (Hendrycks et al., 2021), a fixed 128-question LiveCodeBench v6 subset (Jain et al., 2024), IFBench strict (300 questions) (Pyatkin et al., 2025), and GPQA Diamond (198 questions) (Rein et al., 2023). Mathematics, code, and instruction following use one greedy response per question. GPQA uses four responses per question at temperature 1 and top- $p = 1$. Its score is the mean correctness over all four responses, then over questions: average@4 accuracy. Evaluation permits up to 32,768 response tokens, bounded by the remaining configured 32,768-token context. Each complete suite contains 1,720 responses.

Table 4 gives all thirteen complete suites. Nine suites have local response-level artifacts: the shared initial student, seven multi-teacher checkpoints, and S-PG at update 500. Their 15,480 response records were checked against artifact hashes, expected sample counts, unique prompt/sample indices, and logged aggregate scores. The code artifacts report zero sandbox infrastructure errors. Recomputing all 7,128 GPQA responses with the recorded `final-answer-v2` scorer gives zero score changes. Four single-teacher suites contribute 6,880 responses through previously verified completion summaries from a second node. Their per-question records remain on that node.

The four RL teachers were evaluated separately on their respective domain benchmarks with the compatible native prompt and decoding protocol (Table 5). The science teacher’s 792 saved re-

Table 4: All completed aligned capability scores (%). Scores use the standard dataset metrics and all evaluated responses. The shared initial student is counted once in the thirteen-suite total.

Configuration	Updates	Math	Code	IF	GPQA
Initial	0	68.80	16.41	18.33	32.20
S-PG	100	77.40	18.75	18.67	31.82
S-PG	250	73.80	20.31	17.00	35.35
S-PG	500	73.80	19.53	18.00	31.82
S-I64	100	74.60	18.75	17.67	33.21
S-I64	250	74.40	21.09	18.33	32.45
M-PG	50	69.60	16.41	18.67	28.03
M-PG	100	72.00	17.19	19.33	31.19
M-I64-DR	50	69.60	17.19	19.00	31.69
M-I64-DR	100	71.60	17.97	19.00	32.70
M-I64-DT	50	72.20	18.75	17.00	32.32
M-I64-GT	50	74.20	17.97	17.00	32.95
M-I64-GT	100	74.40	17.97	21.00	32.07

Table 5: Domain teacher reference evaluations, retained from the compatible September 6 evaluation campaign. Science uses GPQA average@4.

Teacher / target domain	Responses	Score (%)	Truncation (%)
Math	500	75.60	6.80
Code	128	23.44	24.22
If	300	29.00	7.33
Science	792	34.60	0.51

sponses also agree exactly with the current GPQA scorer. These teacher scores describe the available supervision on each target domain.

A.3 PAIRED COMPARISONS AND EVIDENCE PROVENANCE

For the paired comparisons, question identity includes the prompt, label, and question metadata, with post-generation response and sandbox diagnostics removed. Correspondence is verified before pairing. We bootstrap the per-question score differences 10,000 times using seed 1042; each GPQA question retains its four-response cluster. Figure 6 shows percentile 95% intervals. These are nominal, unadjusted intervals over evaluation questions, conditional on the trained checkpoints. Training-seed variation is a separate source of uncertainty. The same procedure supplies the initial-to-checkpoint comparisons in the evidence bundle.

The source bundle `experiments/aligned_evidence_20260910` freezes the plotted measurements, prompt-level score records, source SHA-256 hashes, scoring implementation, protocol, and local diagnostic code. Figure generation reads only this bundle and runs on CPU. The manifest identifies the source artifact for each measurement. The aligned series uses the common initial parent and native response semantics throughout; earlier Base initialization, Student64-to-I64 continuations, and superseded teacher-top- k runs are outside this series.

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and domain weights λ_d . Any PG coefficients are detached. For $n_d = |\mathcal{B}_d|$, let $\bar{T}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} T_i$ and $\bar{g}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} g_i$. Define the empirical covariance using the same $1/n_d$ normalization:

$$\text{Cov}_d(T, g) = \frac{1}{n_d} \sum_{i \in \mathcal{B}_d} (T_i - \bar{T}_d)(g_i - \bar{g}_d). \quad (5)$$

Expanding gives $n_d^{-1} \sum_i T_i g_i - \bar{T}_d \bar{g}_d$. Dividing by $\bar{T}_d$ yields Equation 4. GT combines the resulting domain gradients using observed token shares; DT combines them using λ_d ; DR combines $\bar{g}_d$ using

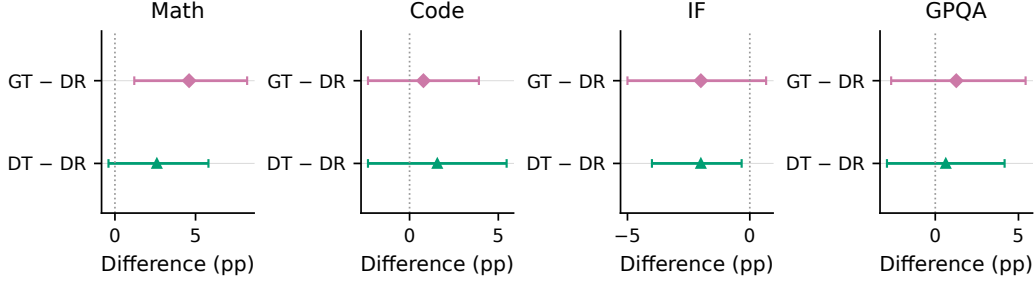


Figure 6: All paired update-50 I64 reduction contrasts against DR. Points give score differences in percentage points and bars give nominal 95% paired prompt-bootstrap intervals. GPQA resamples questions with their four responses together.

λ_d . Multiplying each domain’s globally averaged terms by its target weight divided by its observed token share therefore gives DT exactly.

The derivation is a finite-batch weighted-average identity. Response sampling, masks, and lengths are held fixed. Its population analogue involves an expectation of a ratio; replacing that expectation by a ratio of expectations requires additional assumptions or a large-batch limit.

C PARAMETER AND OPTIMIZER MEASUREMENTS

C.1 COORDINATES AND RECORDED QUANTITIES

The online record distinguishes raw aggregate gradients, clipped gradients, actual FP32 optimizer steps, cumulative FP32 master changes, and cumulative BF16 model changes. Cumulative changes use the same initialization reference. All global statistics use unique model coordinates, excluding vocabulary padding and shared duplicates. Per-layer statistics and absolute-threshold sensitivities are retained in the frozen measurements. The sparsity value is the fraction at or below a threshold; figures report its complement. Energy concentration uses squared Euclidean norm and ranks coordinates by absolute change.

For local calculations, the exported checkpoint contains FP32 master values, BF16 model values, first and second moments, optimizer hyperparameters, and step clocks. The BF16 values match all 1,720,574,976 coordinates in the native exported model. Teacher weight values are verified in the same precision. Local forward/backward computations evaluate these values in FP32 with TF32 disabled. All proposed steps are computed from the same saved state with a non-fused Adam implementation. The online training optimizer uses its own fused implementation.

We evaluate three local states. *Saved* retains both moments and the step clock. *Reset* zeros the first moment while retaining the second moment and clock. *Fresh* zeros both moments and resets the clock. Global norm clipping at 1 is applied before the Adam calculation, and weight decay is zero. For the zero-gradient branch, the saved first moment continues to decay and move the master weights. A no-step master-to-BF16 cast gives an exact match to the saved model. Thirteen numerical checks cover the Adam formulas against PyTorch, loss gradients, parameter-coordinate inner products, support ties, and zero-vector handling. Figure 7 shows how absolute thresholds change the measured local activity.

C.2 RESPONSE BANKS, OVERLAPS, AND GRADIENT MAGNITUDES

Banks use seeds 42 and 43, one response per domain per bank, a 2,048-token prompt cap, and a 256-token generation cap. All eight responses reach the cap. Four response positions are sampled without replacement per response. Each row’s four prefixes receive equal weight. For routed gradients, each teacher scores its domain row; for common-bank gradients, it scores all four rows with equal domain weights. Crossed comparisons score every domain row with every teacher, keeping the number of

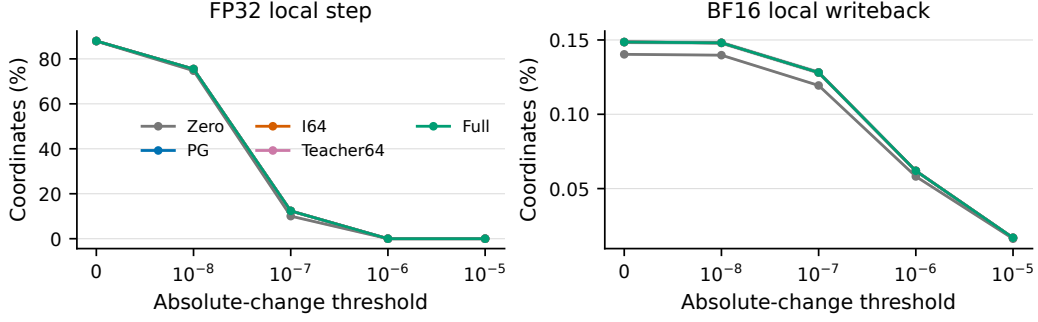


Figure 7: Absolute-threshold sensitivity of the saved-state local steps. Lines show means over two banks and shading shows their range; the zero-gradient control has one value. The two panels use the same thresholds and separate FP32 master changes from BF16 writeback.

prefixes per gradient at four. The 192 crossed-pair records and 48 routed/common pair records share underlying gradients, so points represent paired conditions within banks.

Top- k coordinate sets rank absolute gradient values, using a stable coordinate index to break cutoff ties. The diagnostic records the cutoff and whether a tie crosses the selection boundary. These tie flags accompany the source metrics, including overlaps from nearly degenerate gradients. Table 6 supplies the routed PG norms; the first bank’s instruction-following gradient has norm 1.53×10^{-9} . Cosine normalizes magnitude, so opposite directions for such a row carry very little absolute gradient mass. Jaccard overlap measures selected locations and cosine measures signed direction. For reference, independent uniform sets of fraction ρ have expected-intersection to expected-union ratio $\rho/(2 - \rho)$, or approximately 0.00503 at $\rho = 1\%$. This reference uses uniform coordinates; layer and weight-magnitude structure remain in the measured gradients.

D VOCABULARY COVERAGE AND SAMPLING

The corrected teacher-top-64 diagnostic uses the fixed teacher set T and full-vocabulary probabilities:

$$\ell_{T64} = \sum_{v \in T} \left[p_{\theta}(v) \log \frac{p_{\theta}(v)}{q_d(v)} - p_{\theta}(v) + q_d(v) \right]. \quad (6)$$

Its gradient is $\sum_{v \in T} p_{\theta}(v) \log(p_{\theta}(v)/q_d(v)) \nabla_{\theta} \log p_{\theta}(v)$. I64 instead uses the detached weights in Equation 3, normalized over the original student set S . T64 and full vocabulary are local objective comparisons at the M-PG checkpoint; online capability curves compare PG and I64.

For a fixed prefix, differentiating Equation 2 gives

$$\nabla_{\theta} \ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_{\theta}(v) \left(\log \frac{p_{\theta}(v)}{q_d(v)} + 1 \right) \nabla_{\theta} \log p_{\theta}(v) \quad (7)$$

$$= \mathbb{E}_{a \sim p_{\theta}} \left[\log \frac{p_{\theta}(a)}{q_d(a)} \nabla_{\theta} \log p_{\theta}(a) \right], \quad (8)$$

using $\sum_v \nabla_{\theta} p_{\theta}(v) = 0$. The sampling experiment draws fresh actions from this fixed-prefix distribution with fixed prefix weights. At each sample count, two independent draws are evaluated for each bank; counts use separate, non-nested draws. Relative error is $\|\hat{g} - g_{\text{full}}\|_2 / \|g_{\text{full}}\|_2$. This identity conditions on prefixes and weights. Original rollout losses additionally use the realized response and its eventual length-dependent weight.

E TEACHER DISTRIBUTION DISTANCE

Teacher distance is computed on the full next-token vocabulary at matched prefixes:

$$\text{JS}(q_d, q_e) = \frac{1}{2} D_{\text{KL}}(q_d \| m) + \frac{1}{2} D_{\text{KL}}(q_e \| m), \quad m = \frac{1}{2}(q_d + q_e). \quad (9)$$

Table 6: Fixed-prefix coverage and routed gradient scale. Probability mass and full-vocabulary student entropy (nats) are averaged over each row’s four prefixes. A displayed mass of 100% may reflect rounding at the reported precision.

Bank	Domain	Mass in T (%)	Mass in I (%)	Entropy	$\ g_{PG}\ _2$
1	Math	100.0000	100.0000	0.3243	132.6
1	Code	99.9968	99.9965	0.7701	163.8
1	IF	100.0000	100.0000	1.85×10^{-5}	6.13×10^{-9}
1	Science	99.9184	99.9162	0.7405	54.82
2	Math	99.9991	99.9991	0.3221	2.605
2	Code	100.0000	100.0000	0.2962	3.167
2	IF	100.0000	100.0000	0.5004	55.52
2	Science	99.9993	99.9992	0.2442	0.8505

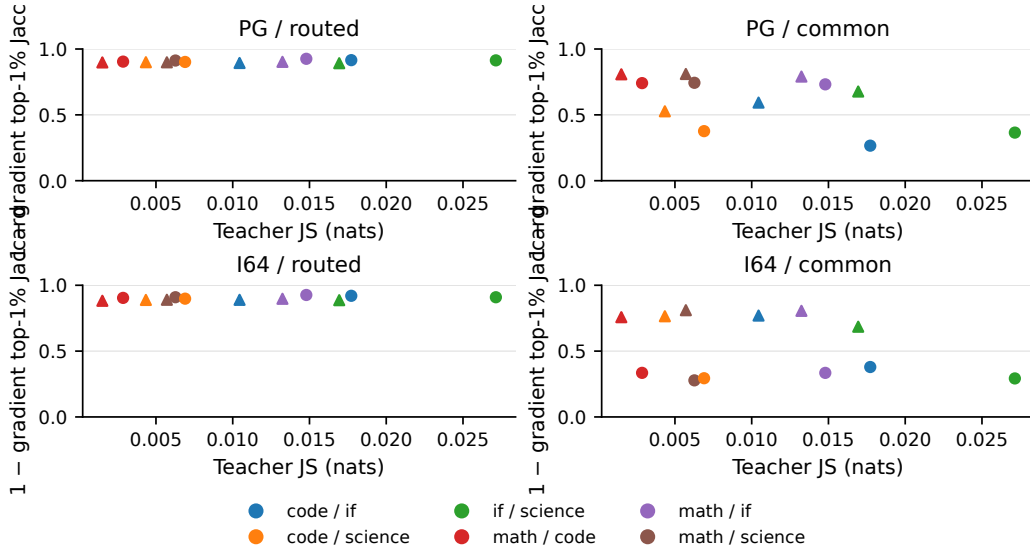


Figure 8: Full-vocabulary teacher JS versus gradient-support distance. Colors identify teacher pairs; circles and triangles identify banks. Each panel pairs the same twelve JS measurements with its PG or I64, routed or common-bank gradient overlap.

We average over the same equally weighted domains and prefixes for every pair. The twelve bank-pair measurements are paired with the corresponding gradient overlaps in Figure 8. Common-bank gradients exhibit a wider overlap range than routed gradients. The two banks and six pairs describe the relationship for these four teachers; comparisons sharing a teacher also share its distribution and gradients.

F RESPONSE BEHAVIOR DURING TRAINING AND EVALUATION

Figure 9 shows all 128 M-PG rollout records, without smoothing. Equal prompt quotas coexist with larger mathematics token shares because mathematics responses are longer. Instruction-following truncation varies over the trajectory. The terminal-token panel records mean teacher-minus-student log probability only at observed completed terminal positions. It measures teacher supervision on those positions, with the corresponding observed-response counts retained in the source records.

Figure 10 shows token shares for the three I64 reductions in the frozen scalar record. Records at the same rollout clock are merged across metric writes before plotting. The frozen histories end at rollout indices 127 for M-PG, 99 for DR and DT, and 114 for GT. These histories record available rollout behavior; capability availability follows the completed suites in Table 3.

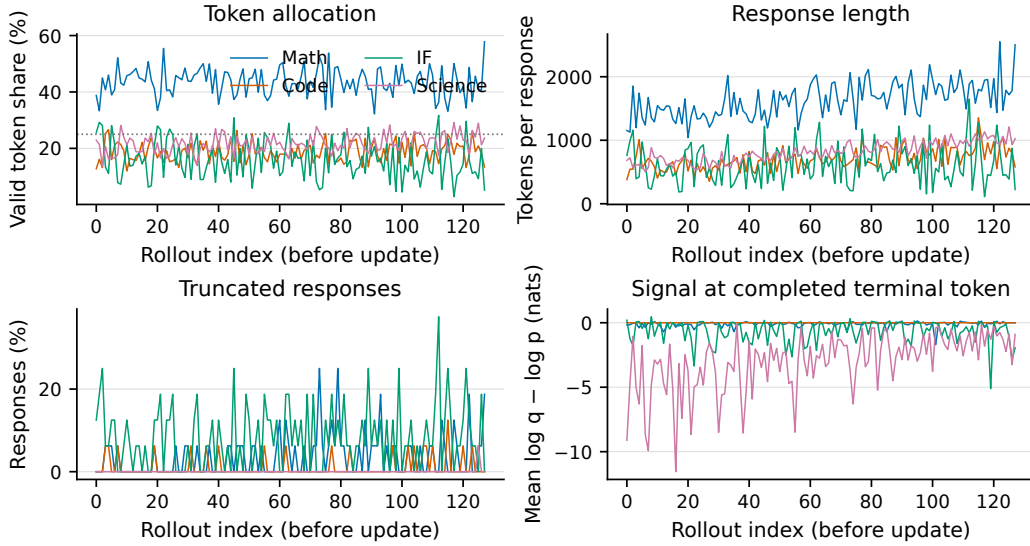


Figure 9: M-PG training response behavior in the available 128-update record. The dotted token-share reference is 25%, matching the prompt quota. Truncation uses the 4,096-token training cap. Terminal signals average completed responses with valid teacher scores. Rollout index r precedes optimizer update $r + 1$.

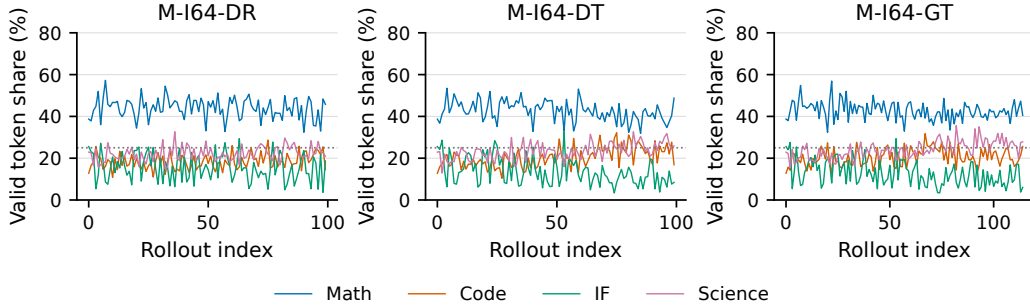


Figure 10: Valid-token allocation under the three I64 reductions. Colors match Figure 9: mathematics, code, instruction following, and science. All points are raw recorded values. The dotted reference is the common 25% prompt share.

Evaluation diagnostics complement the standard scores in Figure 11. The repetition flag applies to responses longer than 10,000 characters whose final 10,000 characters have a UTF-8 byte-length to level-9 zlib compressed-length ratio above 10. Completed, nonrepetitive reward is averaged over all responses, assigning zero to responses outside that category. The standard benchmark score retains its original definition and denominator. The figure uses the shared initial student and update-100 evaluations; the single-teacher summaries contain truncation and repetition but omit the combined reward diagnostic.

G RL TEACHER REGULARIZATION AND DOMAIN WEIGHTS

Our experiments distill RL-trained Qwen3-1.7B teachers into a Qwen3-1.7B student (Appendix A). Teacher training introduces another source of domain weighting: its reward scale and KL penalty determine the signal passed to the student. We derive this connection by treating each teacher as the optimum of its KL-regularized RL objective and distilling its complete response distribution with reverse KL.

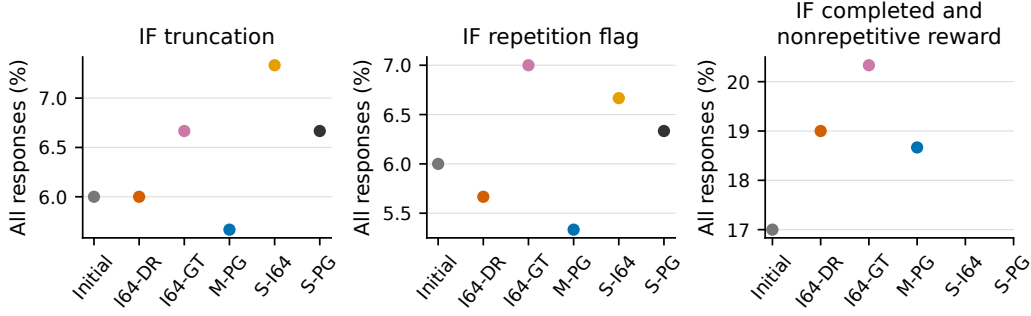


Figure 11: IFBench generation diagnostics alongside the standard capability results. The first two panels show truncation and repetition flags; the third shows reward from completed, nonrepetitive responses divided by the total response count. Missing single-teacher values in the third panel reflect summary availability. Evaluation uses the context-limited 32,768-token cap.

The teacher’s RL objective. During RL, teacher d is rewarded for solving its task and penalized for deviating from a fixed reference policy μ_d . The student starts distillation from π_{θ_0} , which in our experiments is Qwen3-1.7B. Thus μ_d specifies the reference in the teacher’s training objective, while π_{θ_0} specifies the student’s starting checkpoint. For each domain d , fix a prompt distribution $\mathcal{D}_d$, a sequence reward $R_d(x, y)$, and a KL penalty coefficient $\beta_d > 0$. Define

$$\begin{aligned} \mathcal{R}_d(\theta) &= \mathbb{E}_{x \sim \mathcal{D}_d, y \sim \pi_{\theta}(\cdot | x)} [R_d(x, y)], \\ K_d(\theta) &= \mathbb{E}_{x \sim \mathcal{D}_d} D_{\text{KL}}(\pi_{\theta}(\cdot | x) \| \mu_d(\cdot | x)), \quad J_d(\theta) = \mathcal{R}_d(\theta) - \beta_d K_d(\theta). \end{aligned} \quad (10)$$

With common support and finite expectations and normalizers, the optimum over response distributions has the standard form (Rafailov et al., 2023, Appendix A.1):

$$q_d^*(y | x) = \frac{\mu_d(y | x) \exp(R_d(x, y)/\beta_d)}{Z_d(x)}, \quad Z_d(x) = \sum_y \mu_d(y | x) \exp(R_d(x, y)/\beta_d). \quad (11)$$

Gradient equivalence. Consider the complete-sequence reverse KL, without response-length normalization, $L_d^{\text{seq}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_d} D_{\text{KL}}(\pi_{\theta}(\cdot | x) \| q_d^*(\cdot | x))$. Substituting Equation 11 gives

$$\begin{aligned} L_d^{\text{seq}}(\theta) &= K_d(\theta) - \frac{\mathcal{R}_d(\theta)}{\beta_d} + C_d = -\frac{J_d(\theta)}{\beta_d} + C_d, \\ -\nabla_{\theta} L_d^{\text{seq}}(\theta) &= \frac{1}{\beta_d} \nabla_{\theta} J_d(\theta), \quad C_d = \mathbb{E}_{x \sim \mathcal{D}_d} \log Z_d(x). \end{aligned} \quad (12)$$

Since C_d is constant with respect to θ , distilling this teacher has the same gradient direction as maximizing its RL objective, scaled by $1/\beta_d$. The derivative includes the change in the student’s response distribution as its parameters change. The equality holds throughout student training, for any initialization.

How teacher training changes domain weights. With domain weights λ_d , the mixed descent direction separates into a reward term and a reference term:

$$-\nabla_{\theta} \sum_d \lambda_d L_d^{\text{seq}}(\theta) = \sum_d \frac{\lambda_d}{\beta_d} \nabla_{\theta} \mathcal{R}_d(\theta) - \sum_d \lambda_d \nabla_{\theta} K_d(\theta). \quad (13)$$

The reward terms push the student toward higher task rewards; the reference terms pull it toward the teachers’ RL references. The coefficient of domain d ’s reward gradient is therefore λ_d/β_d . For example, with equal domain weights and comparable reward units, mathematics and coding teachers with $\beta_d = 0.05$ and 0.10 give the mathematics reward gradient twice the coefficient of the coding reward gradient. The resulting update also depends on the magnitudes and directions of these gradients and the reference terms.

If $R_d = s_d \tilde{R}_d$ for $s_d > 0$, the coefficient of $\nabla_{\theta} \mathbb{E}[\tilde{R}_d]$ becomes $\lambda_d s_d / \beta_d$. Thus reward scale and KL strength act together: rescaling both by the same positive constant leaves the optimal teacher unchanged.

What student initialization changes. When the student starts from the same policy as every teacher’s RL reference, $\pi_{\theta_0} = \mu_d$, each KL term is minimized and $\nabla_{\theta} K_d(\theta_0) = 0$. The initial descent direction then contains only the reward gradients:

$$-\nabla_{\theta} \sum_d \lambda_d L_d^{\text{seq}}(\theta) \Big|_{\theta_0} = \sum_d \frac{\lambda_d}{\beta_d} \nabla_{\theta} \mathcal{R}_d(\theta) \Big|_{\theta_0}. \quad (14)$$

When the student’s initial policy differs from a teacher’s RL reference, Equation 13 retains the corresponding KL gradient from the first update. Distillation then combines improving task reward with reducing the student’s distance from that reference.

Connection to token balancing. The averaging rules in Section 3 set the weights assigned to domains and responses. The derivation above explains how the teacher’s RL objective sets the reward scale within a domain’s sequence loss. Our token losses in Section 2 use fixed sampled prefixes, response-length averaging, and top- k supervision or clipped advantages. Their local gradients depend on these loss choices as well as the teacher distribution; Equation 12 describes the full derivative of the complete-sequence objective.